Complete Biostatistics Reference

Measurement Error Framework for Spatial Omics Inference

Comprehensive Mathematical Formulations, Derivations, and Examples

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1. Probability Distributions for Count Data

Spatial transcriptomics data consists of *counts* — the number of detected mRNA transcripts per gene per cell. The choice of distributional model fundamentally affects all downstream inference.

1.1 The Poisson Distribution

Definition 1.1 (Poisson Distribution)

A random variable Y follows a Poisson distribution with rate parameter $\lambda > 0$, written $Y \sim \text{Poisson}(\lambda)$, if:

$$P(Y = k) = \frac{\lambda^k e^{-\lambda}}{k!}, \quad k = 0, 1, 2, \ldots$$

Properties:

- **Mean:** $\mathbb{E}[Y] = \lambda$
- **Variance:** $\text{Var}(Y) = \lambda$ (mean equals variance)
- **Mode:** $\lfloor \lambda \rfloor$ (greatest integer $\leq \lambda$)
- **Moment generating function:** $M_Y(t) = \exp(\lambda(e^t - 1))$
- **Cumulant generating function:** $K_Y(t) = \lambda(e^t - 1)$, so all cumulants equal λ
- **Additivity:** If $Y_1 \sim \text{Poisson}(\lambda_1)$ and $Y_2 \sim \text{Poisson}(\lambda_2)$ are independent, then $Y_1 + Y_2 \sim \text{Poisson}(\lambda_1 + \lambda_2)$
- **Thinning:** If $Y \sim \text{Poisson}(\lambda)$ and each event is independently retained with probability δ, the retained count follows $\text{Poisson}(\delta\lambda)$

Example 1.1 — Transcript Detection

A cell expresses gene *EGFR* at rate $\lambda = 8$ transcripts. The platform has detection efficiency $\delta = 0.6$. The observed count is:

$$Y \sim \text{Poisson}(0.6 \times 8) = \text{Poisson}(4.8)$$

Expected count: 4.8. Standard deviation: $\sqrt{4.8} \approx 2.19$. Relative noise: $2.19/4.8 \approx 46\%$.

Probability of observing exactly 3 transcripts:

$$P(Y = 3) = \frac{4.8^3\, e^{-4.8}}{3!} = \frac{110.59 \times 0.00823}{6} = 0.152$$

1.2 The Negative Binomial Distribution

Definition 1.2 (Negative Binomial Distribution)

$Y \sim \mathrm{NB}(\mu, \theta)$ where $\mu > 0$ is the mean and $\theta > 0$ is the dispersion parameter:

$$P(Y = k) = \binom{k + \theta - 1}{k} \left(\frac{\theta}{\theta + \mu}\right)^{\theta} \left(\frac{\mu}{\theta + \mu}\right)^{k}$$

Properties:

- **Mean:** $\mathbb{E}[Y] = \mu$
- **Variance:** $\mathrm{Var}(Y) = \mu + \mu^2/\theta$ (variance > mean, i.e., *overdispersion*)
- **Limit:** As $\theta \to \infty$, $\mathrm{NB}(\mu, \theta) \to \mathrm{Poisson}(\mu)$

Remark 1.1 — Poisson vs. Negative Binomial for Spatial Omics

The Poisson model assumes variance equals mean (technical noise only). The Negative Binomial allows overdispersion (biological variability between cells of the same type). For **imaging-based** spatial transcriptomics (Xenium, MERFISH, CosMx), individual molecule detection is well-modeled by Poisson. For **sequencing-based** platforms (Visium), the NB model is more appropriate due to PCR amplification noise.

In this framework, we use the Poisson model because our primary data type (Xenium) measures individual molecules without amplification. The extension to NB is straightforward by replacing $\mathrm{Var}(Y) = \lambda$ with $\mathrm{Var}(Y) = \lambda + \lambda^2/\theta$.

1.3 The Zero-Inflated Poisson (ZIP) Model

Definition 1.3 (Zero-Inflated Poisson)

$Y \sim \mathrm{ZIP}(\lambda, \pi)$ is a mixture:

$$P(Y = k) = \begin{cases} \pi + (1 - \pi)e^{-\lambda} & k = 0 \\ (1 - \pi)\frac{\lambda^k e^{-\lambda}}{k!} & k \geq 1 \end{cases}$$

where $\pi \in [0, 1)$ is the zero-inflation probability (probability of a "structural" zero — the gene is truly silent in that cell, not just undetected).

Properties:

- $\mathbb{E}[Y] = (1 - \pi)\lambda$
- $\mathrm{Var}(Y) = (1 - \pi)\lambda(1 + \pi\lambda)$

Critical Issue: Biological vs. Technical Zeros

Hafiz et al. (Briefings in Bioinformatics, 2026) showed that all tested imputation methods **overimpute** — they fill in both technical dropouts (good) and biological zeros (bad), destroying biological signal. This makes the distinction between π (biological silence) and detection failure (δ_g) critical for reliable downstream analysis.

1.4 The Multinomial and Dirichlet-Multinomial

For compositional analysis (e.g., cell type proportions in a spatial spot):

Definition 1.4 (Multinomial Distribution)

$\mathbf{Y} = (Y_1, \ldots, Y_T) \sim \text{Multinomial}(n, \mathbf{p})$ where n is total count and $\mathbf{p} = (p_1, \ldots, p_T)$ with $\sum_t p_t = 1$:

$$P(\mathbf{Y} = \mathbf{y}) = \frac{n!}{\prod_t y_t!} \prod_t p_t^{y_t}$$

The **Dirichlet-Multinomial** adds cell type proportion uncertainty: $\mathbf{p} \sim \text{Dirichlet}(\boldsymbol{\alpha})$, then $\mathbf{Y} \mid \mathbf{p} \sim \text{Multinomial}(n, \mathbf{p})$.

2. The Poisson Measurement Model for Spatial Omics

2.1 Model Specification

THE COMPLETE OBSERVATION MODEL

For cell $i \in \{1, \ldots, N\}$ at spatial position $\mathbf{s}_i \in \mathbb{R}^2$ and gene $g \in \{1, \ldots, G\}$:

$$y_{ig} \sim \text{Poisson}(\delta_g \cdot \lambda_{ig}), \quad g \in S$$
$$y_{ig} = \text{unobserved}, \quad g \notin S$$

where:

- $y_{ig} \in \mathbb{Z}_+ = \{0, 1, 2, \ldots\}$ is the observed transcript count
- $\lambda_{ig} > 0$ is the true biological expression rate
- $\delta_g \in (0, 1]$ is the gene-specific detection efficiency
- $S \subset \{1, \ldots, G\}$ is the measured gene panel with $|S| = m \ll G$

2.2 The Selection Matrix Formulation

Define the **selection matrix** $A_S \in \{0, 1\}^{m \times G}$:

$$(A_S)_{jg} = \begin{cases} 1 & \text{if gene } g \text{ is the } j\text{-th element of panel } S \\ 0 & \text{otherwise} \end{cases}$$

A_S consists of m rows of the $G \times G$ identity matrix. The observed data for cell i:

$$\mathbf{y}_{i,\text{obs}} = A_S \, \mathbf{y}_i \in \mathbb{Z}_+^m$$

with each component satisfying:

$$[A_S \mathbf{y}_i]_j \sim \text{Poisson}([A_S \, \boldsymbol{\delta} \odot \boldsymbol{\lambda}_i]_j)$$

where $\odot$ is the Hadamard (elementwise) product.

2.3 Signal-to-Noise Analysis

SNR DECOMPOSITION

The signal-to-noise ratio for gene g in cell i:

$$\mathrm{SNR}_{ig} = \frac{\mathbb{E}[y_{ig}]^2}{\mathrm{Var}(y_{ig})} = \frac{(\delta_g \lambda_{ig})^2}{\delta_g \lambda_{ig}} = \delta_g \lambda_{ig}$$

The **coefficient of variation**:

$$\mathrm{CV}_{ig} = \frac{\sqrt{\mathrm{Var}(y_{ig})}}{\mathbb{E}[y_{ig}]} = \frac{1}{\sqrt{\delta_g \lambda_{ig}}}$$

Example 2.1 — SNR Across Expression Levels

With detection efficiency $\delta = 0.7$:

Gene Type	λ (true rate)	$\delta\lambda$ (observed rate)	SNR	CV	Quality
Housekeeping (GAPDH)	200	140	140	8.5%	Excellent
Marker (CD3E)	30	21	21	21.8%	Good
Signaling (CXCL12)	5	3.5	3.5	53.5%	Poor
Rare TF (FOXP3)	0.5	0.35	0.35	169%	Unreliable

Many ligands and receptors critical for CCI analysis (CXCL12, CCL5, etc.) fall in the **poor SNR regime**, making CCI inference inherently fragile even before imputation.

2.4 The Full Data Matrix

$$\mathbf{Y} \in \mathbb{Z}_+^{N \times G} \quad \text{(full count matrix — mostly unobserved)}$$

$$\mathbf{Y}_{\text{obs}} = \mathbf{Y} A_S^\top \in \mathbb{Z}_+^{N \times m} \quad \text{(observed data)}$$

$$\mathbf{\Lambda} \in \mathbb{R}_+^{N \times G} \quad \text{(true rate matrix — the target)}$$

The **observation fraction**: $\rho = m/G$.

Platform	Panel Size m	ρ	Resolution

Xenium (standard)	313	1.6%	Subcellular
Xenium 5K	5,000	25%	Subcellular
CosMx 6K	6,000	30%	Subcellular
MERFISH	100–500	0.5–2.5%	Subcellular
Visium	~18,000	~90%	Multi-cell spot
Visium HD	~18,000	~90%	2μm bins

2.5 Log-Likelihood Function

The joint log-likelihood of all observed data, given the rate matrix $\mathbf{\Lambda}$:

POISSON LOG-LIKELIHOOD

$$\ell(\mathbf{\Lambda}; \mathbf{Y}_{\text{obs}}) = \sum_{i=1}^{N} \sum_{g \in S} [y_{ig} \log(\delta_g \lambda_{ig}) - \delta_g \lambda_{ig} - \log(y_{ig}!)]$$

Dropping constant terms and absorbing δ_g into λ_{ig}:

$$\ell(\mathbf{\Lambda}) = \sum_{i=1}^{N} \sum_{g \in S} [y_{ig} \log \lambda_{ig} - \lambda_{ig}]$$

3. Low-Rank Matrix Structure and Gene Programs

3.1 The Biological Basis for Low Rank

Gene expression is not arbitrary. Cells activate coordinated *gene programs* — sets of genes regulated together by shared transcription factors, signaling pathways, or epigenetic states. For example:

- **Proliferation program:** MKI67, TOP2A, CDK1, PCNA co-express
- **Interferon response:** ISG15, MX1, IFIT1, STAT1 co-express
- **EMT program:** VIM, FN1, SNAI1/2, ZEB1 co-express

If there are K independent programs, the expression of all G genes is determined by K latent factors — the rate matrix $\mathbf{\Lambda}$ is approximately rank-K.

3.2 The Log-Linear Factor Model

GENE PROGRAM DECOMPOSITION

$$\log \mathbf{\Lambda} = \mathbf{1}\boldsymbol{\mu}^\top + \mathbf{U}\mathbf{V}^\top + \mathbf{E}$$

Elementwise:

$$\log \lambda_{ig} = \mu_g + \sum_{k=1}^{K} u_{ik} v_{gk} + \epsilon_{ig}$$

Symbol	Dimension	Interpretation	What Determines It
μ_g	scalar per gene	Baseline expression level of gene g	Gene identity (housekeeping vs. rare)
u_{ik}	scalar per cell-program pair	Activity of program k in cell i	Cell type + cell state
v_{gk}	scalar per gene-program pair	Membership of gene g in program k	Gene regulatory network
ϵ_{ig}	scalar per cell-gene pair	Residual noise	Stochastic gene expression

3.3 Identifiability and Parameter Counting

Total unknowns in the factor model (ignoring $\mathbf{E}$):

$$\text{Parameters} = \underbrace{G}_{\boldsymbol{\mu}} + \underbrace{NK}_{\mathbf{U}} + \underbrace{GK}_{\mathbf{V}} = G(1+K) + NK$$

Observations: Nm (one count per cell per measured gene).

Proposition 3.1 (Necessary Condition for Identifiability)

Recovery of $\mathbf{\Lambda}$ from $\mathbf{Y}_{\text{obs}}$ requires at minimum:

$$Nm \geq G(1+K) + NK$$

$$\Leftrightarrow \quad m \geq \frac{G(1+K)}{N} + K$$

For $N \gg G$: $m \gtrsim K$.

For $N = G$: $m \gtrsim 2K + 1$.

Example 3.1 — Minimum Panel Size

Scenario	N	G	K	Min m
Xenium breast cancer	100,000	20,000	30	~37
Small tissue section	5,000	20,000	30	~150
Rare cell study	500	20,000	30	~1,230

Key insight: **more cells help**. Large datasets can tolerate smaller panels because there is more information to estimate the gene programs.

3.4 Empirical Rank Estimation

To estimate K empirically from data $\mathbf{Y}_{\text{obs}}$:

1. Normalize: $\tilde{\mathbf{Y}} = \log(1 + \text{CP10K}(\mathbf{Y}_{\text{obs}}))$
2. Center: $\bar{\mathbf{Y}} = \tilde{\mathbf{Y}} - \mathbf{1}\bar{\mathbf{y}}^\top$
3. Compute SVD: $\bar{\mathbf{Y}} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top$
4. K = number of singular values σ_k exceeding the Marchenko-Pastur threshold (random matrix theory baseline)

The **Marchenko-Pastur law** gives the expected largest singular value of a random $N \times m$ matrix with i.i.d. entries of variance σ^2:

$$\sigma_{\text{MP}} = \sigma\left(1 + \sqrt{N/m}\right)$$

Singular values exceeding this are signal; those below are noise.

4. Matrix Completion and Recovery Theory

4.1 The Recovery Problem

Given observed entries $\{y_{ig} : g \in S\}$ and the low-rank model, we seek to recover the full rate matrix $\mathbf{\Lambda} \in \mathbb{R}_+^{N \times G}$.

PENALIZED POISSON MAXIMUM LIKELIHOOD

$$\hat{\mathbf{\Lambda}} = \arg\min_{\mathbf{\Lambda} \succ 0} \left[-\ell(\mathbf{\Lambda}) + \tau \|\mathbf{\Lambda}\|_*\right]$$

where $\|\mathbf{\Lambda}\|_* = \sum_k \sigma_k(\mathbf{\Lambda})$ is the nuclear norm (convex relaxation of rank), and $\tau > 0$ is the regularization strength.

4.2 The Incoherence Condition

Definition 4.1 (Incoherence)

A rank-K matrix $\mathbf{\Lambda} \in \mathbb{R}^{N \times G}$ with SVD $\mathbf{\Lambda} = \mathbf{U\Sigma V}^\top$ is μ_0-**incoherent** if:

$$\max_i \|\mathbf{U}_{i,:}\|^2 \leq \frac{\mu_0 K}{N}, \qquad \max_g \|\mathbf{V}_{g,:}\|^2 \leq \frac{\mu_0 K}{G}$$

This requires that no single cell or gene dominates the signal. $\mu_0 = 1$ is the best case (uniform spread); $\mu_0 = N/K$ is the worst (all signal in one cell).

Biological interpretation: Incoherence fails when one cell type is extremely rare (all signal in a few cells) or one gene uniquely defines a program (all signal in one gene). This predicts that **rare cell types are harder to recover** — a biologically meaningful limitation.

4.3 Recovery Guarantee

Theorem 4.1 (Poisson Matrix Completion; after Cao & Xie, 2015)

Let $\mathbf{\Lambda} \in \mathbb{R}_+^{N \times G}$ have rank K, be μ_0-incoherent, with $\lambda_{\max} = \max_{i,g} \lambda_{ig}$. If the panel S is chosen uniformly at random with $|S| = m$, and:

$$m \geq C_0 \mu_0^2 K^2 \log^2(N + G)$$

then with probability at least $1 - (N + G)^{-3}$:

$$\frac{1}{NG} \|\hat{\mathbf{\Lambda}} - \mathbf{\Lambda}\|_F^2 \leq C_1 \cdot \frac{K(N + G)\lambda_{\max}}{Nm}$$

For $N \gg G$, this simplifies to:

$$\boxed{\text{MSE} \approx C \cdot \frac{K\lambda_{\max}}{m}}$$

4.4 Understanding the Bound

Factor	Effect on MSE	Why
K increases	MSE increases linearly	More latent dimensions to estimate from same number of measurements
m increases	MSE decreases as $1/m$	More observed genes provide more constraints

$\lambda_{\max}$ increases	MSE increases linearly	Poisson variance scales with rate; highly expressed genes contribute more noise
N increases	MSE decreases as $1/N$	More cells share the same gene programs, providing more samples to estimate $\mathbf{V}$

4.5 The Factored Estimator

In practice, we solve the factored form (computationally more efficient than nuclear norm minimization):

FACTORED POISSON NMF

$$\hat{\mathbf{U}}, \hat{\mathbf{V}} = \arg\min_{\mathbf{U},\mathbf{V}} \sum_{i=1}^{N} \sum_{g \in S} \left[\exp\left(\mu_g + \mathbf{u}_i^\top \mathbf{v}_g\right) - y_{ig}\left(\mu_g + \mathbf{u}_i^\top \mathbf{v}_g\right)\right] + \frac{\tau}{2}\left(\|\mathbf{U}\|_F^2 + \|\mathbf{V}\|_F^2\right)$$

The regularization $\|\mathbf{U}\|_F^2 + \|\mathbf{V}\|_F^2$ implicitly constrains the nuclear norm of $\mathbf{U}\mathbf{V}^\top$.

5. Per-Gene Reconstruction Error Analysis

5.1 MSE Decomposition

The reconstruction error for gene g can be decomposed into two independent sources:

PER-GENE MSE

$$\mathrm{MSE}(g) = \underbrace{\frac{1}{\delta_g \bar{\lambda}_g}}_{\substack{\text{Poisson shot noise} \\ \text{(irreducible)}}} + \underbrace{\|\mathbf{v}_g\|^2 (1 - R_g^2)}_{\substack{\text{Panel incompleteness} \\ \text{(reducible by panel design)}}}$$

where $\bar{\lambda}_g = \frac{1}{N}\sum_i \lambda_{ig}$ is the mean expression of gene g.

5.2 Gene Predictability: The R_g^2 Statistic

Definition 5.1 (Latent-Space R^2)

The predictability of unmeasured gene g from panel S, in the latent factor space:

$$R_g^2 = \frac{\mathbf{v}_g^\top \mathbf{P}_{V_S} \mathbf{v}_g}{\|\mathbf{v}_g\|^2}$$

where $\mathbf{P}_{V_S} = \mathbf{V}_S^\top (\mathbf{V}_S \mathbf{V}_S^\top)^{-1} \mathbf{V}_S$ is the projection matrix onto the column space of $\mathbf{V}_S^\top$ (the latent representations of the measured genes).

Derivation: The expression of gene g in the latent space is $\mathbf{v}_g \in \mathbb{R}^K$. The best linear predictor of $\mathbf{v}_g$ from the measured gene loadings $\mathbf{V}_S = [\mathbf{v}_{g'}]_{g' \in S}$ is the orthogonal projection $\hat{\mathbf{v}}_g = \mathbf{P}_{V_S} \mathbf{v}_g$. The fraction of variance explained is R_g^2.

Residual variance:

$$\sigma_g^2 = \|\mathbf{v}_g - \hat{\mathbf{v}}_g\|^2 = \|\mathbf{v}_g\|^2 - \mathbf{v}_g^\top \mathbf{P}_{V_S} \mathbf{v}_g = \|\mathbf{v}_g\|^2 (1 - R_g^2)$$

Example 5.1 — Gene Predictability

Suppose $K = 3$ programs. Gene *CXCL12* has loading $\mathbf{v}_{\text{CXCL12}} = (0.8, 0.3, 0.1)$ (primarily in program 1). If the panel S contains genes that span programs 1 and 2 but not 3:

- The projection captures the (0.8, 0.3) components but misses 0.1
- $R^2_{\text{CXCL12}} = (0.8^2 + 0.3^2)/(0.8^2 + 0.3^2 + 0.1^2) = 0.73/0.74 = 0.986$
- Residual variance: $0.74 \times (1 - 0.986) = 0.010$
- This gene is highly predictable — imputation will work well.

Now consider gene *FOXP3* with loading $\mathbf{v}_{\text{FOXP3}} = (0.05, 0.05, 0.9)$ (primarily in the uncovered program 3):

- $R^2_{\text{FOXP3}} = (0.05^2 + 0.05^2)/(0.05^2 + 0.05^2 + 0.9^2) = 0.005/0.815 = 0.006$
- Residual variance: $0.815 \times (1 - 0.006) = 0.810$
- This gene is **essentially unrecoverable** from this panel.

6. Spatial Autocorrelation: Moran's I

6.1 Definition and Formula

MORAN'S I (GLOBAL SPATIAL AUTOCORRELATION)

Question: "Are nearby cells more similar in expression than expected by chance?"

$$I = \frac{N}{W} \cdot \frac{\sum_{i=1}^{N}\sum_{j=1}^{N} w_{ij}(x_i - \bar{x})(x_j - \bar{x})}{\sum_{i=1}^{N}(x_i - \bar{x})^2}$$

Equivalently, using matrix notation with $\mathbf{x} = (x_1, \ldots, x_N)^\top$, centered $\tilde{\mathbf{x}} = \mathbf{x} - \bar{x}\mathbf{1}$:

$$I = \frac{N}{W} \cdot \frac{\tilde{\mathbf{x}}^\top \mathbf{W} \tilde{\mathbf{x}}}{\tilde{\mathbf{x}}^\top \tilde{\mathbf{x}}}$$

where $\mathbf{W}$ is the spatial weight matrix with $W_{ij} = w_{ij}$ and $W = \sum_{ij} w_{ij}$.

6.2 Expected Value and Variance Under the Null

Proposition 6.1 (Null Distribution of Moran's I)

Under the null hypothesis of no spatial autocorrelation (random permutation of values across locations):

$$\mathbb{E}[I] = -\frac{1}{N-1}$$

$$\mathrm{Var}(I) = \frac{N^2 S_1 - N S_2 + 3W^2}{W^2(N^2-1)} - \left(\frac{1}{N-1}\right)^2$$

where $S_1 = \frac{1}{2}\sum_i \sum_j (w_{ij} + w_{ji})^2$ and $S_2 = \sum_i (w_{i\cdot} + w_{\cdot i})^2$.

The **Z-score** for testing spatial autocorrelation:

$$Z_I = \frac{I - \mathbb{E}[I]}{\sqrt{\mathrm{Var}(I)}}$$

Under the null, $Z_I \xrightarrow{d} \mathcal{N}(0,1)$ for large N.

6.3 Complete Worked Example

Example 6.1 — 9-Cell Grid, Gene X Expression

Cells in a 3×3 grid with expression values and neighbor connections (horizontal and vertical adjacency):

```
Position:  (1,1) (1,2) (1,3)     Expression:  8  7  9
           (2,1) (2,2) (2,3)                  7  8  2
           (3,1) (3,2) (3,3)                  1  2  1
```

$N = 9$ cells. 12 undirected edges (= 24 directed pairs). $W = 24$.

Step 1: Compute mean.

$$\bar{x} = \frac{8+7+9+7+8+2+1+2+1}{9} = \frac{45}{9} = 5.0$$

Step 2: Compute deviations.

Cell i	x_i	$x_i - \bar{x}$	$(x_i - \bar{x})^2$
1	8	+3	9
2	7	+2	4
3	9	+4	16
4	7	+2	4
5	8	+3	9
6	2	-3	9
7	1	-4	16
8	2	-3	9
9	1	-4	16

$$\sum_i (x_i - \bar{x})^2 = 9 + 4 + 16 + 4 + 9 + 9 + 16 + 9 + 16 = 92$$

Step 3: Compute cross-products for each neighbor pair.

Edge	$x_i - \bar{x}$	$x_j - \bar{x}$	**Product**
1↔2	+3	+2	+6
1↔4	+3	+2	+6
2↔3	+2	+4	+8

2↔5	+2	+3	+6
3↔6	+4	-3	-12
4↔5	+2	+3	+6
4↔7	+2	-4	-8
5↔6	+3	-3	-9
5↔8	+3	-3	-9
6↔9	-3	-4	+12
7↔8	-4	-3	+12
8↔9	-3	-4	+12

Sum of products (each edge counted once): $6 + 6 + 8 + 6 - 12 + 6 - 8 - 9 - 9 + 12 + 12 + 12 = 30$

Double sum (symmetric w_{ij}): $\sum_i \sum_j w_{ij}(x_i - \bar{x})(x_j - \bar{x}) = 2 \times 30 = 60$

Step 4: Compute Moran's I.

$$I = \frac{9}{24} \cdot \frac{60}{92} = 0.375 \times 0.652 = \mathbf{0.245}$$

Step 5: Significance test.

$$\mathbb{E}[I] = -\frac{1}{9-1} = -0.125$$

Under normality approximation with $\mathrm{Var}(I) \approx 0.035$ (computed from S_1, S_2 formulas):

$$Z_I = \frac{0.245 - (-0.125)}{\sqrt{0.035}} = \frac{0.370}{0.187} = 1.98$$

Two-sided $p = 0.048$ — marginally significant spatial autocorrelation.

6.4 Interpretation Scale

I Value	Z_I	Interpretation	Spatial Pattern
$+0.8$ to $+1.0$	$\gg 2$	Strong positive autocorrelation	Smooth gradient or large domains
$+0.3$ to $+0.8$	> 2	Moderate positive autocorrelation	Spatial clustering with mixing zones
-0.125 to $+0.3$	≈ 0	No significant spatial pattern	Random distribution

-0.5 to -0.125	< -2	Negative autocorrelation	Alternating / checkerboard

6.5 Local Moran's I (LISA)

Definition 6.1 (Local Indicator of Spatial Association)

$$I_i = \frac{(x_i - \bar{x})}{\frac{1}{N}\sum_j (x_j - \bar{x})^2} \sum_j w_{ij}(x_j - \bar{x})$$

Properties: $\sum_i I_i = N \cdot I$ (local values sum to global). Local Moran's I_i identifies individual cells contributing to spatial autocorrelation: **HH** (hot spot), **LL** (cold spot), **HL** (spatial outlier high), **LH** (spatial outlier low).

7. Spatial Autocorrelation: Geary's C

7.1 Definition and Formula

GEARY'S C (LOCAL DISSIMILARITY MEASURE)

Question: "How different are neighboring cells from each other?"

$$C = \frac{(N-1)}{2W} \cdot \frac{\sum_{i=1}^{N}\sum_{j=1}^{N} w_{ij}(x_i - x_j)^2}{\sum_{i=1}^{N}(x_i - \bar{x})^2}$$

7.2 Relationship to Moran's I

Expanding $(x_i - x_j)^2 = (x_i - \bar{x})^2 - 2(x_i - \bar{x})(x_j - \bar{x}) + (x_j - \bar{x})^2$:

$$C = \frac{(N-1)}{2W} \cdot \frac{\sum_{ij} w_{ij}\left[(x_i - \bar{x})^2 + (x_j - \bar{x})^2\right] - 2\sum_{ij} w_{ij}(x_i - \bar{x})(x_j - \bar{x})}{\sum_i (x_i - \bar{x})^2}$$

For a symmetric weight matrix with constant row sums (regular lattice): $C \approx 1 - I$. In general, C and I are inversely but not perfectly linearly related.

7.3 Null Distribution

Proposition 7.1

Under random permutation:

$$\mathbb{E}[C] = 1, \qquad Z_C = \frac{C - 1}{\sqrt{\mathrm{Var}(C)}}$$

7.4 Comparison Table

Property	Moran's I	Geary's C
Numerator uses	$(x_i - \bar{x})(x_j - \bar{x})$ — cross-products relative to *global* mean	$(x_i - x_j)^2$ — *direct* pairwise squared differences
Sensitivity	Global pattern (overall mean-based)	Local variation (edge-level)
Clustering	$I \to +1$	$C \to 0$
Random	$I \to -1/(N-1) \approx 0$	$C \to 1$
Dispersion	$I \to -1$	$C \to 2$
Robust to outliers	Less (products can be dominated by extreme deviations from mean)	More (squared differences are local)

7.5 Worked Example (Same 9-Cell Grid)

Example 7.1 — Geary's C Computation

Edge	x_i	x_j	$(x_i - x_j)^2$	Location
1↔2	8	7	1	Within high region
1↔4	8	7	1	Within high region
2↔3	7	9	4	Within high region
2↔5	7	8	1	Within high region
3↔6	9	2	**49**	**Boundary**
4↔5	7	8	1	Within high region
4↔7	7	1	**36**	**Boundary**
5↔6	8	2	**36**	**Boundary**
5↔8	8	2	**36**	**Boundary**
6↔9	2	1	1	Within low region
7↔8	1	2	1	Within low region
8↔9	2	1	1	Within low region

Sum of squared differences (single-counted): $1 + 1 + 4 + 1 + 49 + 1 + 36 + 36 + 36 + 1 + 1 + 1 = 168$

Double sum: $2 \times 168 = 336$

$$C = \frac{(9-1)}{2 \times 24} \cdot \frac{336}{92} = \frac{8}{48} \times 3.652 = 0.167 \times 3.652 = \mathbf{0.609}$$

Key observation: The 4 boundary edges (3↔6, 4↔7, 5↔6, 5↔8) contribute $49 + 36 + 36 + 36 = 157$ out of 168 total — **93.5%** of the signal. Geary's C is dominated by the boundary between high and low expression regions.

8. Neighborhood Enrichment (NE) Score

8.1 Definition

NEIGHBORHOOD ENRICHMENT SCORE

Question: "Do cells of type a neighbor cells of type b more (or less) often than expected by chance?"

For cell types a and b:

$$NE(a,b) = \frac{n_{\text{obs}}(a,b) - \mu_{\text{perm}}(a,b)}{\sigma_{\text{perm}}(a,b)}$$

where:

- $n_{\text{obs}}(a,b) = \sum_{(i,j)\in\mathcal{E}} \mathbb{1}[c_i = a, c_j = b]$ (observed count of a-b neighbor pairs)
- $\mu_{\text{perm}}(a,b) = \mathbb{E}_\pi[n(a,b)]$ (expected count under random label permutations)
- $\sigma_{\text{perm}}(a,b) = \text{Std}_\pi[n(a,b)]$ (standard deviation under permutations)

8.2 Analytical Expected Value

Under random permutation of cell type labels, the expected number of a-b edges:

$$\mu_{\text{perm}}(a,b) = |\mathcal{E}| \cdot \begin{cases} \frac{n_a(n_a-1)}{N(N-1)} & \text{if } a = b \\ \frac{2\,n_a\,n_b}{N(N-1)} & \text{if } a \neq b \end{cases}$$

where $n_a = |\{i : c_i = a\}|$ is the number of cells of type a, and $|\mathcal{E}|$ is the total number of directed edges.

8.3 Complete Worked Example

Example 8.1 — Tumor-Fibroblast Neighborhood Enrichment

9 cells in a 3×3 grid: 5 Tumor (T) cells, 4 Fibroblast (F) cells.

```
Cell types:     T  T  T
                T  T  F
                F  F  F
```

12 undirected edges. Edges by type pair:

Edge	Types	Pair
1↔2	T↔T	T-T
1↔4	T↔T	T-T
2↔3	T↔T	T-T
2↔5	T↔T	T-T
3↔6	T↔F	T-F
4↔5	T↔T	T-T
4↔7	T↔F	T-F
5↔6	T↔F	T-F
5↔8	T↔F	T-F
6↔9	F↔F	F-F
7↔8	F↔F	F-F
8↔9	F↔F	F-F

Observed counts: T-T = 5, T-F = 4, F-F = 3.

Expected T-T count under random permutation ($n_T = 5, N = 9$, directed edges $= 24$):

$$\mu(T,T) = 24 \times \frac{5 \times 4}{9 \times 8} = 24 \times \frac{20}{72} = 24 \times 0.278 = 6.67 \text{ directed} = 3.33 \text{ undirected}$$

From 10,000 permutations, $\sigma(T,T) \approx 1.2$.

$$NE(T,T) = \frac{5 - 3.33}{1.2} = \frac{1.67}{1.2} = \mathbf{+1.39}$$

$NE(T,T) > 0$: tumor cells are **enriched** as neighbors of other tumor cells (spatial clustering).

Expected T-F count:

$$\mu(T,F) = 24 \times \frac{2 \times 5 \times 4}{9 \times 8} = 24 \times \frac{40}{72} = 13.33 \text{ directed} = 6.67 \text{ undirected}$$

$$NE(T,F) = \frac{4 - 6.67}{\sigma(T,F)} < 0$$

$NE(T,F) < 0$: tumor and fibroblast cells are **depleted** as neighbors (they tend to segregate).

8.4 NE Score Matrix

For T cell types, the full NE matrix is $T \times T$:

$$\mathbf{NE} = \begin{pmatrix} NE(1,1) & NE(1,2) & \cdots & NE(1,T) \\ NE(2,1) & NE(2,2) & \cdots & NE(2,T) \\ \vdots & & \ddots & \vdots \\ NE(T,1) & NE(T,2) & \cdots & NE(T,T) \end{pmatrix}$$

Diagonal entries: self-clustering. Off-diagonal: cross-type interactions. This matrix is the primary output of squidpy's `sq.gr.nhood_enrichment()`.

9. Cell-Cell Interaction (CCI) Test Statistic

9.1 The Ligand-Receptor Co-Expression Score

CCI TEST STATISTIC

For ligand gene l and receptor gene r, the spatial co-expression score:

$$T_{lr} = \frac{1}{|\mathcal{E}|} \sum_{(i,j) \in \mathcal{E}} x_{il} \cdot x_{jr}$$

where $\mathcal{E} = \{(i,j) : w_{ij} = 1\}$ is the set of spatial neighbor pairs.

Biological interpretation: T_{lr} measures whether cells expressing ligand l tend to be spatially adjacent to cells expressing receptor r. High T_{lr} suggests a potential signaling interaction.

9.2 Variants

Variant	Formula	Used By
Product score	$\frac{1}{\lvert\mathcal{E}\rvert} \sum_{(i,j)} x_{il} x_{jr}$	CellPhoneDB, squidpy

Correlation-based	$\frac{\sum_{(i,j)}(x_{il}-\bar{x}_l)(x_{jr}-\bar{x}_r)}{\lvert\mathcal{E}\rvert\cdot\sigma_l\sigma_r}$	SpatialDM
Binary co-expression	$\frac{1}{\lvert\mathcal{E}\rvert}\sum_{(i,j)}\mathbb{1}[x_{il}>0]\cdot\mathbb{1}[x_{jr}>0]$	CellChat (simplified)
Weighted by distance	$\sum_{(i,j)} k(\lVert\mathbf{s}_i-\mathbf{s}_j\rVert)\, x_{il}\, x_{jr}$	COMMOT

9.3 Null Model and Expected Value

Under the null of no spatial interaction (expression and location are independent):

$$\mathbb{E}_0[T_{lr}] = \bar{x}_l \cdot \bar{x}_r + \frac{I_W}{N-1}(\bar{x^2}_l - \bar{x}_l^2)^{1/2}(\bar{x^2}_r - \bar{x}_r^2)^{1/2}$$

where I_W is the Moran-like autocorrelation of the spatial weight matrix. For most spatial graphs, $I_W/(N-1)$ is negligible, so:

$$\mathbb{E}_0[T_{lr}] \approx \bar{x}_l \cdot \bar{x}_r$$

10. Permutation Testing and Multiple Comparison Correction

10.1 Permutation Test for CCI

ALGORITHM: PERMUTATION TEST FOR CCI SIGNIFICANCE

1. Compute observed statistic: T_{lr}^{obs}
2. For $b = 1, \ldots, B$:
   - Draw random permutation π_b of cell indices
   - Compute null statistic: $T_{lr}^{(b)} = \frac{1}{\lvert\mathcal{E}\rvert}\sum_{(i,j)\in\mathcal{E}} x_{\pi_b(i),l} \cdot x_{j,r}$
3. p-value: $p = \frac{\#\{b:T_{lr}^{(b)} \geq T_{lr}^{\text{obs}}\}+1}{B+1}$

The "+1" in numerator and denominator ensures the p-value is never exactly 0 (conservative correction).

10.2 Benjamini-Hochberg FDR Control

Definition 10.1 (False Discovery Rate)

$$\text{FDR} = \mathbb{E}\left[\frac{V}{R \vee 1}\right]$$

where V = number of false rejections (false positives), R = total rejections, and $a \vee b = \max(a, b)$.

BENJAMINI-HOCHBERG PROCEDURE

Given L tests with p-values $p_{(1)} \leq p_{(2)} \leq \cdots \leq p_{(L)}$:

1. Find the largest k such that $p_{(k)} \leq \frac{k}{L} \cdot q$, where q is the target FDR (typically 0.05)
2. Reject all hypotheses $H_{(1)}, \ldots, H_{(k)}$

Adjusted p-values: $p_{(i)}^{\text{BH}} = \min\left\{\frac{L}{i} p_{(i)},\ 1\right\}$ (with monotonicity enforcement).

10.3 Impact of Multiple Testing on CCI

Example 10.1

Testing 200 LR pairs at FDR $q = 0.05$. If 40 have true signal and 160 are null:

- Expected false discoveries at 5% FDR: $0.05 \times R \approx 0.05 \times 50 = 2.5$
- But if imputation inflates scores, many null pairs cross the significance threshold
- BH procedure assumes valid p-values — biased test statistics produce **anti-conservative** p-values, and BH no longer controls FDR at the nominal level

This is why regression calibration (Section 12) is essential: it restores validity of the p-values.

11. Bias Decomposition Under Measurement Error

11.1 The Measurement Error Model

After imputation, the estimated expression for cell i, gene g is:

$$\hat{\lambda}_{ig} = \lambda_{ig} + e_{ig}$$

where e_{ig} is the imputation error with:

- $\mathbb{E}[e_{ig}] \approx 0$ (unbiased imputation, approximately)
- $\text{Var}(e_{ig}) = \text{MSE}(g)$ (gene-specific error variance)
- $\text{Cov}(e_{ig}, e_{jg}) \neq 0$ when $w_{ij} = 1$ (**spatially correlated errors**)

11.2 Bias in the CCI Statistic

BIAS DERIVATION

The CCI statistic computed on imputed data:

$$\hat{T}_{lr} = \frac{1}{|\mathcal{E}|} \sum_{(i,j)\in\mathcal{E}} \hat{\lambda}_{il} \cdot \hat{\lambda}_{jr}$$

Substituting $\hat{\lambda}_{il} = \lambda_{il} + e_{il}$ and $\hat{\lambda}_{jr} = \lambda_{jr} + e_{jr}$:

$$\begin{aligned} \hat{\lambda}_{il} \cdot \hat{\lambda}_{jr} &= (\lambda_{il} + e_{il})(\lambda_{jr} + e_{jr}) && (1) \\ &= \lambda_{il}\lambda_{jr} + \lambda_{il}e_{jr} + \lambda_{jr}e_{il} + e_{il}e_{jr} && (2) \end{aligned}$$

Taking expectations:

$$\mathbb{E}[\hat{T}_{lr}] = \underbrace{\frac{1}{|\mathcal{E}|} \sum_{(i,j)} \lambda_{il}\lambda_{jr}}_{=T_{lr}^{\text{true}}} + \underbrace{\frac{1}{|\mathcal{E}|} \sum_{(i,j)} \lambda_{il}\mathbb{E}[e_{jr}]}_{=0} + \underbrace{\frac{1}{|\mathcal{E}|} \sum_{(i,j)} \lambda_{jr}\mathbb{E}[e_{il}]}_{=0} + \underbrace{\frac{1}{|\mathcal{E}|} \sum_{(i,j)} \mathbb{E}[e_{il}e_{jr}]}_{=\text{BIAS}} \quad (3)$$

11.3 Why the Bias Is Positive for Graph-Based Methods

The bias term is:

$$\text{bias} = \frac{1}{|\mathcal{E}|} \sum_{(i,j)\in\mathcal{E}} \text{Cov}(e_{il}, e_{jr})$$

The Fundamental Problem

Graph-based imputation methods (GNN, graph diffusion, spatial smoothing) share information across spatial neighbors. If cell j is a neighbor of cell i, the imputed value $\hat{\lambda}_{jg}$ depends partly on $\hat{\lambda}_{ig}$ (through message passing). This means:

$$\text{Cov}(e_{il}, e_{jr}) > 0 \quad \text{when } (i,j) \in \mathcal{E}$$

But $\mathcal{E}$ is the *exact same set of edges* used in the CCI computation. The imputation method creates positive correlations on the same edges where we test for biological interactions. The bias is **structurally guaranteed to be positive**.

11.4 Magnitude Estimation

Under the approximation that error correlations are proportional to Moran's I:

$$\frac{1}{|\mathcal{E}|} \sum_{(i,j)\in\mathcal{E}} \text{Cov}(e_{il}, e_{jr}) \approx \rho_{lr} \cdot \sqrt{\text{MSE}(l) \cdot \text{MSE}(r)}$$

where $\rho_{lr} = \sqrt{|I_l| \cdot |I_r|}$ is the cross-gene spatial error correlation and I_l, I_r are Moran's I values of the imputed expression for the ligand and receptor genes.

12. Regression Calibration for CCI Correction

Regression calibration (Carroll, Ruppert, Stefanski & Crainiceanu, 2006) is a standard technique for correcting bias in statistics computed from error-contaminated variables.

12.1 The Correction

REGRESSION-CALIBRATED CCI STATISTIC

$$\hat{T}_{lr}^{\text{corrected}} = \hat{T}_{lr}^{\text{naive}} - \widehat{\text{bias}}_{lr}$$

where:

$$\widehat{\text{bias}}_{lr} = \hat{\rho}_{lr} \cdot \sqrt{\widehat{\text{MSE}}(l) \cdot \widehat{\text{MSE}}(r)}$$

$$\hat{\rho}_{lr} = \sqrt{|I_l| \cdot |I_r|}$$

12.2 MSE Estimation via Cross-Validation

For **measured genes** ($g \in S$): direct comparison with observed counts.

$$\widehat{\text{MSE}}(g) = \frac{1}{N} \sum_{i=1}^{N} (x_{ig}^{\text{norm}} - \hat{\lambda}_{ig})^2, \quad g \in S$$

For **unmeasured genes** ($g \notin S$): predict MSE from a meta-model trained on measured genes. Features: mean expression, variance, latent-space R_g^2.

$$\widehat{\text{MSE}}(g) = f_{\text{meta}}(\bar{\hat{\lambda}}_g,\ \text{Var}(\hat{\lambda}_{ig}),\ R_g^2), \quad g \notin S$$

12.3 Corrected Variance and P-Values

CORRECTED INFERENCE

The variance of the corrected statistic includes uncertainty from the correction itself:

$$\text{Var}(\hat{T}_{lr}^{\text{corr}}) \approx \text{Var}(\hat{T}_{lr}^{\text{naive}}) + \frac{\widehat{\text{MSE}}(l) \cdot \widehat{\text{MSE}}(r)}{|\mathcal{E}|}$$

The corrected permutation test subtracts the same bias estimate from null replicates:

$$T_{lr}^{(b),\text{corr}} = T_{lr}^{(b),\text{naive}} - \widehat{\text{bias}}_{lr}$$

$$p^{\text{corr}} = \frac{\#\{b : T_{lr}^{(b),\text{corr}} \geq \hat{T}_{lr}^{\text{corr}}\} + 1}{B + 1}$$

13. Error Propagation to Cell Type Classification

13.1 Nearest-Centroid Classifier

Cell type classification assigns cell i to the type whose centroid is closest in expression space:

$$\hat{c}_i = \arg\min_t \|\hat{\boldsymbol{\lambda}}_i - \boldsymbol{\mu}^{(t)}\|_2$$

13.2 Misclassification Bound

Theorem 13.1 (Error Propagation to Cell Typing)

For a cell of true type t, the misclassification probability satisfies:

$$P(\hat{c}_i \neq t) \leq \exp\left(-\frac{\Delta_t^2}{8\,\sigma_{\text{recon}}^2}\right)$$

where:

$$\Delta_t = \min_{t' \neq t} \|\boldsymbol{\mu}^{(t)} - \boldsymbol{\mu}^{(t')}\|_2$$

$$\sigma_{\text{recon}}^2 = \sum_{g=1}^{G} \text{MSE}(g)$$

Derivation sketch: Misclassification occurs when $\|\hat{\boldsymbol{\lambda}}_i - \boldsymbol{\mu}^{(t')}\| < \|\hat{\boldsymbol{\lambda}}_i - \boldsymbol{\mu}^{(t)}\|$ for some $t' \neq t$. This requires the error $\mathbf{e}_i = \hat{\boldsymbol{\lambda}}_i - \boldsymbol{\lambda}_i$ to have a component of magnitude $\geq \Delta_t/2$ along the direction $\boldsymbol{\mu}^{(t)} - \boldsymbol{\mu}^{(t')}$. By the sub-Gaussian tail bound on the projected error: $P(\langle \mathbf{e}_i, \mathbf{d}_{tt'} \rangle > \Delta_t/2) \leq \exp(-\Delta_t^2/(8\sigma_{\text{recon}}^2))$.

13.3 Critical Panel Size per Cell Type

Setting the misclassification probability to a target α:

$$m_t^* = \frac{8\,C\,K\,\lambda_{\max}}{\Delta_t^2 \cdot \log(1/\alpha)}$$

Example 13.1

With $K = 30$, $\lambda_{\max} = 500$, $C = 2$, target $\alpha = 0.05$:

Cell Type Pair	Δ_t	m_t^*	Identifiable at $m = 300$?
Tumor vs Fibroblast	45.2	26	Yes
Macrophage vs Monocyte	12.8	325	Marginal
CD4 naive vs CD4 memory	4.1	3,162	No

14. Error Propagation to CCI False Positive Rate

CCI VARIANCE INFLATION

$$\mathrm{Var}(\hat{T}_{lr}) \geq \frac{1}{|\mathcal{E}|}\left[\bar{\lambda}_r^2 \cdot \mathrm{MSE}(l) + \bar{\lambda}_l^2 \cdot \mathrm{MSE}(r) + \mathrm{MSE}(l) \cdot \mathrm{MSE}(r)\right]$$

Scenario	Dominant Variance Term	Scaling
Both measured ($l, r \in S$)	Poisson noise only	$O(1/\delta\bar{\lambda})$
One imputed ($l \notin S, r \in S$)	$\bar{\lambda}_r^2 \cdot \mathrm{MSE}(l)$	Linear in MSE
Both imputed ($l, r \notin S$)	$\mathrm{MSE}(l) \cdot \mathrm{MSE}(r)$	**Quadratic** in MSE

Quadratic Degradation

When both ligand and receptor are imputed, the variance of the CCI test statistic grows as the *product* of their individual MSEs. This means CCI reliability degrades quadratically — far faster than gene-level imputation quality alone would suggest.

15. Error Propagation to Spatial Domain Detection

Spatial domain detection clusters cells based on expression similarity + spatial proximity. The effective similarity between neighboring cells:

$$w_{ij}^{\mathrm{eff}} = \mathrm{sim}(\hat{\boldsymbol{\lambda}}_i, \hat{\boldsymbol{\lambda}}_j) \cdot f(\|\mathbf{s}_i - \mathbf{s}_j\|)$$

Reconstruction error perturbs similarity:

$$\mathrm{sim}(\hat{\boldsymbol{\lambda}}_i, \hat{\boldsymbol{\lambda}}_j) \approx \mathrm{sim}(\boldsymbol{\lambda}_i, \boldsymbol{\lambda}_j) + O\left(\frac{\sigma_{\mathrm{recon}}}{\Delta_{\min}}\right)$$

Domains **merge** (under-segmentation) or **split** (over-segmentation) when $\sigma_{\mathrm{recon}}/\Delta_{\min} \gtrsim 1$, where $\Delta_{\min}$ is the minimum inter-domain expression distance.

16. Panel Perturbation Stability Analysis

16.1 Procedure

PANEL PERTURBATION ALGORITHM

Input: Panel S (m genes), imputation function f, downstream analysis ϕ.

1. For $b = 1, \ldots, B$:
   - Sample sub-panel $S_b^* \subset S$ by dropping fraction α of genes uniformly at random
   - Re-impute: $\hat{\mathbf{\Lambda}}_b = f(\mathbf{Y}_{\mathrm{obs}}, S_b^*)$
   - Re-run downstream: $\theta_b = \phi(\hat{\mathbf{\Lambda}}_b)$ (e.g., CCI score for each LR pair)
2. Compute stability metrics from $\{\theta_1, \ldots, \theta_B\}$

16.2 Stability Metrics

Metric	Formula	Interpretation
Coefficient of Variation	$\mathrm{CV} = \mathrm{Std}(\theta_b)/\lvert\mathrm{Mean}(\theta_b)\rvert$	Relative variability of the conclusion
Stability Score	$\mathrm{Stab} = 1 - \mathrm{clip}(\mathrm{CV}, 0, 1)$	1 = robust, 0 = unstable
Sign Consistency	$\mathrm{SC} = \lvert\mathrm{Mean}(\mathrm{sign}(\theta_b))\rvert$	1 = always same sign, 0 = random sign
Significance Fraction	$\mathrm{SF} = \mathrm{Mean}(\mathbb{1}[p_b < 0.05])$	Fraction of sub-panels yielding significance

16.3 Why This Is Not a Bootstrap

Remark 16.1 — Methodological Honesty

A bootstrap resamples *observations* (cells) and provides coverage guarantees under regularity conditions. Our procedure resamples *features* (genes) — a fundamentally different operation. We **do not claim** that the perturbation intervals have nominal coverage. Instead, we claim:

1. Stability (low CV across sub-panels) **predicts** actual reliability
2. Instability (high CV) is a **diagnostic** flag for unreliable conclusions

This must be validated empirically by showing correlation between predicted stability and actual error (ground truth comparison).

17. Calibration and Discrimination Metrics

17.1 Discrimination (Ranking Quality)

Can we rank LR pairs from most to least reliable?

$$r_s = \text{Spearman}(\text{Stability}_{lr},\ |\hat{T}_{lr} - T_{lr}^{\text{true}}|)$$

Target: $|r_s| \geq 0.7$. A negative r_s means high stability predicts low error (desired).

17.2 Calibration (Absolute Accuracy)

Does predicted reliability match actual false discovery proportion?

CALIBRATION PLOT CONSTRUCTION

1. Restrict to LR pairs called significant in imputed data
2. Bin by predicted reliability (quintiles)
3. In each bin b, compute:

$$\text{FDP}(b) = 1 - \frac{|\{(l,r) \in b : (l,r) \in \text{True Significant}\}|}{|\{(l,r) \in b\}|}$$

4. Plot FDP vs. mean reliability per bin

Perfect calibration: FDP decreases monotonically as reliability increases. Highest reliability bin should have FDP < 0.1; lowest reliability bin should have FDP > 0.5.

17.3 Expected Calibration Error (ECE)

$$\text{ECE} = \sum_{b=1}^{B} \frac{|n_b|}{n} \left| \text{FDP}(b) - (1 - \overline{\text{Stab}}(b)) \right|$$

where n_b is the number of pairs in bin b and $\overline{\text{Stab}}(b)$ is the mean stability in bin b. Lower ECE = better calibration.

18. Gene Panel Optimization Theory

18.1 Optimal Panel Selection as Subset Selection

Given the framework, the optimal panel S^* minimizes expected downstream error:

$$S^* = \arg \min_{S \subset \{1,\ldots,G\}, |S|=m} \sum_{(l,r)\in\mathcal{L}} \text{Var}(\hat{T}_{lr} \mid S)$$

where $\mathcal{L}$ is the set of LR pairs of interest. This is a combinatorial optimization problem (NP-hard in general).

18.2 Greedy Approximation

Using the submodularity of R_g^2 as a function of the panel S (adding a gene always reduces residual variance, but with diminishing returns), a greedy algorithm achieves a $(1 - 1/e) \approx 63\%$ approximation ratio:

GREEDY PANEL DESIGN

1. Initialize $S = \emptyset$
2. For $j = 1, \ldots, m$:

$$g_j^* = \arg\max_{g \notin S} \sum_{(l,r)\in\mathcal{L}} \left[R_l^2(S \cup \{g\}) - R_l^2(S) + R_r^2(S \cup \{g\}) - R_r^2(S)\right]$$

$S \leftarrow S \cup \{g_j^*\}$

19. Fisher Information and Cramér-Rao Bounds

19.1 Fisher Information for Poisson

Definition 19.1

For $Y \sim \text{Poisson}(\lambda)$, the Fisher information:

$$\mathcal{I}(\lambda) = \mathbb{E}\left[\left(\frac{\partial}{\partial\lambda}\log P(Y;\lambda)\right)^2\right] = \frac{1}{\lambda}$$

Derivation:

$$\frac{\partial}{\partial\lambda}\log P(Y;\lambda) = \frac{\partial}{\partial\lambda}[Y\log\lambda - \lambda] = \frac{Y}{\lambda} - 1$$

$$\mathcal{I}(\lambda) = \mathbb{E}\left[\left(\frac{Y}{\lambda} - 1\right)^2\right] = \frac{\mathrm{Var}(Y)}{\lambda^2} = \frac{\lambda}{\lambda^2} = \frac{1}{\lambda}$$

19.2 Cramér-Rao Lower Bound

Theorem 19.1 (Cramér-Rao Bound for Poisson)

For any unbiased estimator $\hat{\lambda}$ of λ based on a single observation $Y \sim \mathrm{Poisson}(\lambda)$:

$$\mathrm{Var}(\hat{\lambda}) \geq \frac{1}{\mathcal{I}(\lambda)} = \lambda$$

For N independent cells observing gene g (all with the same rate λ_g):

$$\mathrm{Var}(\hat{\lambda}_g) \geq \frac{\lambda_g}{N}$$

Remark 19.1

The Cramér-Rao bound for Poisson has a surprising implication: **higher expression is harder to estimate precisely in absolute terms** (variance = λ/N), but **easier to estimate in relative terms** (CV = $1/\sqrt{N\lambda}$). For CCI, where we compute products $\lambda_{il} \cdot \lambda_{jr}$, the absolute error matters more.

19.3 Fisher Information for the Rate Matrix

For the observed entries of $\mathbf{\Lambda}$, the Fisher information matrix:

$$[\mathcal{I}(\mathbf{\Lambda})]_{(i,g),(i',g')} = \begin{cases} \frac{1}{\lambda_{ig}} & \text{if } (i,g) = (i',g') \text{ and } g \in S \\ 0 & \text{otherwise} \end{cases}$$

The total information about the rank-K structure from Nm observations:

$$\text{Total Fisher information} = \sum_{i=1}^{N} \sum_{g \in S} \frac{1}{\lambda_{ig}}$$

20. Variance Estimation and Confidence Intervals

20.1 Variance of the CCI Statistic

For the true (non-imputed) CCI statistic with Poisson-distributed counts:

$$\mathrm{Var}(T_{lr}) = \frac{1}{|\mathcal{E}|^2} \sum_{(i,j) \in \mathcal{E}} \mathrm{Var}(x_{il} \cdot x_{jr})$$

Since x_{il} and x_{jr} are independent (different cells):

$$\mathrm{Var}(x_{il} \cdot x_{jr}) = \mathbb{E}[x_{il}^2]\mathbb{E}[x_{jr}^2] - (\mathbb{E}[x_{il}])^2(\mathbb{E}[x_{jr}])^2$$

$$= (\lambda_{il}^2 + \lambda_{il})(\lambda_{jr}^2 + \lambda_{jr}) - \lambda_{il}^2\lambda_{jr}^2$$

$$= \lambda_{il}^2\lambda_{jr} + \lambda_{il}\lambda_{jr}^2 + \lambda_{il}\lambda_{jr}$$

20.2 Delta Method for Transformed Statistics

For a function $h(\hat{T}_{lr})$ of the CCI statistic (e.g., log-transform for normalization):

$$\mathrm{Var}(h(\hat{T}_{lr})) \approx [h'(T_{lr})]^2 \cdot \mathrm{Var}(\hat{T}_{lr})$$

20.3 Confidence Intervals via Percentile Method

From the panel perturbation distribution $\{\hat{T}_{lr}^{(1)}, \ldots, \hat{T}_{lr}^{(B)}\}$:

$$\mathrm{CI}_{1-\alpha}(\hat{T}_{lr}) = \left[\hat{T}_{lr}^{(\lfloor B\alpha/2 \rfloor)},\ \hat{T}_{lr}^{(\lceil B(1-\alpha/2) \rceil)}\right]$$

where $\hat{T}_{lr}^{(k)}$ is the k-th order statistic. These are *stability intervals*, not confidence intervals with guaranteed coverage.

21. False Discovery Rate Control

21.1 Local FDR

The local false discovery rate for an LR pair with test statistic T:

$$\mathrm{lfdr}(T) = \frac{\pi_0 f_0(T)}{f(T)}$$

where π_0 is the proportion of true nulls, f_0 is the null density, and f is the mixture density.

21.2 Storey's Positive FDR

Definition 21.1

$$\mathrm{pFDR}(t) = \frac{\pi_0 \cdot P_0(T \geq t)}{P(T \geq t)}$$

Estimated $\hat{\pi}_0$ via Storey's method:

$$\hat{\pi}_0 = \frac{\#\{i : p_i > \lambda\}}{L(1-\lambda)}$$

for a tuning parameter $\lambda \in (0, 1)$ (typically $\lambda = 0.5$).

21.3 Impact of Imputation Bias on FDR

FDR Invalidation Under Biased Statistics

BH and Storey procedures assume valid p-values (uniform under the null). When CCI statistics are biased by imputation, the null p-values are **stochastically smaller than uniform** (anti-conservative). This means:

- Nominal FDR control at 5% may produce actual FDR of 20–50%
- Regression calibration restores validity by debiasing the test statistic
- After correction, the permutation p-values should again be approximately uniform under the null

22. Complete Model Summary and Notation

THE ENTIRE FRAMEWORK

Observation:	$y_{ig} \sim \text{Poisson}(\delta_g \lambda_{ig}),\ g \in S,\ \lvert S\rvert = m$
Structure:	$\log \lambda_{ig} = \mu_g + \sum_{k=1}^{K} u_{ik} v_{gk} + \epsilon_{ig}$
Recovery MSE:	$\frac{1}{NG} \lVert \hat{\mathbf{\Lambda}} - \mathbf{\Lambda} \rVert_F^2 \le C \frac{K \lambda_{\max}}{m}$
Per-gene error:	$\text{MSE}(g) = \frac{1}{\delta_g \bar{\lambda}_g} + \lVert \mathbf{v}_g \rVert^2 (1 - R_g^2)$
CCI bias:	$\mathbb{E}[\hat{T}_{lr}] = T_{lr}^{\text{true}} + \rho_{lr} \sqrt{\text{MSE}(l) \cdot \text{MSE}(r)}$
Corrected CCI:	$\hat{T}_{lr}^{\text{corr}} = \hat{T}_{lr} - \hat{\rho}_{lr} \sqrt{\widehat{\text{MSE}}(l) \cdot \widehat{\text{MSE}}(r)}$
Cell typing:	$P(\text{misclassify}) \le \exp\left(-\Delta_i^2 / 8 \sum_g \text{MSE}(g)\right)$
CCI variance:	$\text{Var}(\hat{T}_{lr}) \propto \text{MSE}(l) \cdot \text{MSE}(r)$ (both imputed)

Complete Notation Table

Symbol	Type	Definition	Typical Range
N	Scalar	Number of cells	$10^3 - 10^6$
G	Scalar	Total genes	$\sim 20{,}000$
m	Scalar	Panel size $\lvert S\rvert$	$100 - 5{,}000$
K	Scalar	Latent rank (gene programs)	$15 - 60$
S	Set	Measured gene panel	$\subset \{1, \ldots, G\}$

$\mathbf{s}_i$	$\mathbb{R}^2$	Spatial coordinates of cell i	
y_{ig}	$\mathbb{Z}_+$	Observed transcript count	$0 - 10^3$
λ_{ig}	$\mathbb{R}_+$	True expression rate	$0.01 - 10^3$
$\hat{\lambda}_{ig}$	$\mathbb{R}_+$	Imputed expression	
e_{ig}	$\mathbb{R}$	Imputation error	
δ_g	$(0, 1]$	Detection efficiency	$0.3 - 0.9$
$\boldsymbol{\mu}$	$\mathbb{R}^G$	Baseline log-expression	
$\mathbf{U}$	$\mathbb{R}^{N \times K}$	Cell loading matrix	
$\mathbf{V}$	$\mathbb{R}^{G \times K}$	Gene program matrix	
$\mathbf{v}_g$	$\mathbb{R}^K$	Gene g's program vector	
R_g^2	$[0, 1]$	Gene predictability from S	
w_{ij}	$\{0, 1\}$	Spatial adjacency	
$\mathcal{E}$	Set	Spatial edge set	
T_{lr}	$\mathbb{R}$	CCI test statistic	
I_g	$[-1, 1]$	Moran's I for gene g	
C_g	$[0, 2]$	Geary's C for gene g	
$NE(a, b)$	$\mathbb{R}$	Neighborhood enrichment	
ρ_{lr}	$[0, 1]$	Spatial error correlation	
Δ_t	$\mathbb{R}_+$	Min inter-type distance	
α	$(0, 1)$	Gene drop fraction	$0.1 - 0.3$
B	$\mathbb{Z}_+$	Number of perturbations	$20 - 500$

Complete Biostatistics Reference — Measurement Error Framework for Spatial Omics Inference

22 Sections • All formulations paper-ready with derivations and examples